

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under **220** miles. Rhett drove at an average speed of **35 miles per hour (mph)**, and Jessica drove at an average speed of **40 mph**. Which of the following inequalities represents this situation, where *r* is the number of hours Rhett drove and *j* is the number of hours Jessica drove?

Answer

- A. $35r + 40j > 220$
- B. $35r + 40j < 220$
- C. $40r + 35j > 220$
- D. $40r + 35j < 220$

Correct Answer: B

Rationale

Choice B is correct. It’s given that Rhett drove at an average speed of **35** miles per hour and that he drove for *r* hours. Multiplying **35** miles per hour by *r* hours yields **35*r*** miles, or the distance that Rhett drove. It’s also given that Jessica drove at an average speed of **40** miles per hour and that she drove for *j* hours. Multiplying **40** miles per hour by *j* hours yields **40*j*** miles, or the distance that Jessica drove. The total distance, in miles, that Rhett and Jessica drove can be represented by the expression **35*r* + 40*j***. It’s given that the total distance they drove was under **220** miles. Therefore, the inequality **35*r* + 40*j* < 220** represents this situation.

Choice A is incorrect. This inequality represents a situation in which the total distance Rhett and Jessica drove was over, rather than under, **220** miles.

Choice C is incorrect. This inequality represents a situation in which Rhett drove at an average speed of **40**, rather than **35**, miles per hour, Jessica drove at an average speed of **35**, rather than **40**, miles per hour, and the total distance they drove was over, rather than under, **220** miles.

Choice D is incorrect. This inequality represents a situation in which Rhett drove at an average speed of **40**, rather than **35**, miles per hour, and Jessica drove at an average speed of **35**, rather than **40**, miles per hour.